SAVANT FRAMEWORK

The P.T Field Manual

Engineering and Operating the Forty P.T Entities Across LLM Systems
Machine · Laws · Dynamics · Portability · Evidence · Lineage

Author: Dr. Christabel Odeta

SAVANT FRAMEWORK · CLAI-OS · AEGIS-GLOBAL

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1. Reading Guide & The Core Doctrine

This manual defines the machine that operates the forty P.T entities across heterogeneous LLM systems. It follows the fixed reading order of the corpus: **the machine is defined first, its laws second, the two dynamics problems third, the evidence last.** A reader who reads only Sections 1–3 and Section 7 still holds the irreducible core: the transition system, the operator table, the laws, and the recovery protocol — and still gets the discipline, because every load-bearing claim in this manual carries its status and its kill condition.

> **D1 — The load-bearing distinction.** The model proposes; the machine decides; time and the world constrain both. No generative output enters system state directly. Everything an LLM emits is a proposal; only the deterministic host may commit.

> **D2 — Bounded trust.** The generative model appears in the formalism as an environment process emitting proposals. It receives precisely the amount of trust the architecture places in it — the trust class τ of the entity it is executing — and not one bit more.

> **D3 — Audited shipping.** Nothing ships above its audited state — including the auditing machinery itself. Every claim, verdict, arrest, and release carries an epistemic status and a kill condition; the conformance harness and this manual carry theirs.

> **D4 — Human terminus.** Every authority chain ends at a named human. The machine may propose, detect, arrest below the irreversible rung, and bind; irreversible acts require human enactment (Law 5 of the P.T 51–70 canon; Law D2 of the P.T 71–90 canon).

Status discipline for this document: **ARC** (architecture — implementable specification), **FOR** (formal — derived or definitional), **EMP** (empirical — requires validation), **CON** (conjecture), **OPEN** (explicitly unresolved). Where a claim is EMP, its validation harness is named. The novelty question for the whole construction is declared OPEN in Section 9, not asserted.

2. The Machine

2.1 Definition

The P.T Runtime is a deterministic host M that treats any LLM as an external environment process. Formally (status: FOR — definitional):

```
M = (S, E, A, G, δ, s0)

S   — machine state: the union of all registries (claims, kill criteria, bindings,
      0-sets, freshness certificates, escrow ledger, arrest records, precedents)
E   — the environment process: the LLM, emitting proposals P_gen; opaque to M,
      modeled only by its interface contract and its observed conformance record
A   — operator set: the forty P.T entities, each a partial function
      a_i : S × In_i → Prop_i  (an operator never writes state; it returns a proposal)
G   — the guard: a total validation function Prop → {accept, reject, escalate}
δ   — the transition function: δ(S, p) = S' iff G(p) = accept; otherwise S is
      unchanged (reject) or routed to human review (escalate)
s0  — genesis state: empty registries plus the constitution (Section 3)
```

2.2 The trust classes

Every entity call runs under a declared trust class τ, which caps what its proposal may touch (status: ARC):

Table 1 Trust classes bounding what a proposal may commit

Class	Name	Proposal may	Example entities
τ0	Draft	Return text for human reading only; no state write	P.T 71, 72, 73 analyses
τ1	Registry write	Write to registries after schema + invariant validation	P.T 51, 52, 55, 56
τ2	Trigger	Write plus emit events other operators consume	P.T 53, 54, 66, 76
τ3	Gate proposal	Propose arrests, releases, verdicts — human-gated commit	P.T 77, 84, 89
τ4	Reserved	Never assigned to a model; execution acts only	machine + named humans

2.3 Components

- **Proposal Layer** — model-facing. Builds Entity Call Envelopes (Section 5.1), dispatches to the configured adapter, receives raw proposals.
- **Guard G** — model-free, deterministic, five stages in fixed order: schema validation → invariant check against the laws of Section 3 → epistemic check (status labels present, kill criteria attached where required, freshness in force) → authority check (τ class, human-gate requirements) → commit or route.
- **State Store** — the registries. Append-only where records are evidentiary (P.T 77, 78, 89); versioned where content is living (Θ-sets, term registries).
- **Evidence Bus** — the event spine connecting operators (Section 4.2). Tamper-evident: hash-chained records (lineage: Merkle 1980, Section 8).

- **Human Authority Interface** — the only path to τ4 acts. Presents decision, obligations, reversibility class; then holds the silence (P.T 68).
- **Audit Ledger** — every guard decision, accepted or rejected, with reasons. The ledger is itself claim-bearing state and obeys D3.

2.4 The load-bearing theorem-shaped claim

Claim (FOR, derivable from the definitions): if G is total and δ applies only accepted proposals, then no state transition is attributable to the model — every committed transition passed a deterministic, replayable check. **Corollary (ARC):** model failure modes (drift, sycophancy, hallucination) become proposal-quality problems, not state-integrity problems. **Kill condition:** a demonstrated committed transition whose proposal would fail G as specified — a guard bypass — voids the claim and triggers the recovery protocol (Section 7.4).

3. The Laws (State Invariants)

The laws are not advice to the model; they are invariants checked by G on every proposed transition. I1–I6 restate the six design laws of P.T 51–70; I7–I12 restate the six defense ethics of P.T 71–90; all twelve are machine-checkable or are explicitly marked as requiring human judgment (status: ARC).

Table 2 The twelve invariants and their guard-stage tests

ID	Invariant	Guard test
I1	Surrender to truth — proposals conflicting with established registry state are rejected or routed	Conflict scan against established claims
I2	Registered death — every empirical claim carries an executable kill criterion	Kill-criterion presence and format check
I3	Lawful absence — abstention is always an acceptable proposal; fabrication triggers quarantine	Abstention channel always available; fabrication detector on provenance fields
I4	Bounded language — high-status terms require the definition → observable → measurement chain	Term-registry lookup on flagged vocabulary
I5	Human enactment — publication, authorization, execution require human commit	τ4 gate; no model path to execution
I6	Telos discipline — leverage claims require mechanism and measurement	Leverage-claim schema check (P.T 69)
I7	Defense only — no proposal targets entities outside the governed system	Scope check on all P.T 76–87 outputs

I8	Proportionality — arrests and sanctions select the least incapacitating effective rung	Ladder-order check in P.T 77 proposals
I9	Evidence before verdict — bindings and verdicts require tamper-evident records	Evidence-chain completeness check
I10	The binder is bound — governance outputs carry the same requirements as governed outputs	Self-application test (below)
I11	Truth is the terminal charge — defense actions must not degrade epistemic state	Post-action epistemic delta check
I12	Freshness — expired claims cannot ground new commits	Freshness-certificate validation (P.T 55)

The self-application test (I10), stated plainly: the conformance suite, the guard rules, and this manual are themselves registered as claim-bearing artifacts. They carry provenance (P.T 51), kill criteria (P.T 52), freshness certificates (P.T 55), and are subject to constitutional audit (P.T 70). An auditing machine that exempted itself would void the distinction D3 exists to protect. **Kill condition for I10 as implemented:** discovery of any governance artifact in production without a registry entry.

4. The Two Dynamics Problems

4.1 Problem one — intra-entity dynamics: one entity surviving its context

An LLM call is memoryless; an entity is not. P.T 55's half-lives, P.T 78's escrowed claims, and P.T 76's baselines are long-lived state; the model's context window is short-lived and rented. The engineering consequence (status: ARC): **all entity state lives in the State Store, never in conversation.** The conversation carries only (a) the entity's role text and protocol, (b) the minimal state slice the guard has cleared for this call, and (c) the required output schema. When the window fills or the model version changes, the entity survives because nothing it owns lives in the window.

- **State externalization rule:** any fact the entity will need in a later call must be committed to a registry in this call, or it does not exist.
- **State slice selection:** the Proposal Layer computes the minimal sufficient slice (entity-specific; e.g., P.T 61 receives only the Θ-set for the jurisdictions in scope, not the corpus).
- **Chunking contract:** long inputs are chunked by the host, never by the model; each chunk is tagged so the model's partial outputs can be assembled and validated deterministically.

- **Continuation discipline:** a multi-call entity run carries a run-id; partial proposals are quarantined until the run completes and the assembled proposal passes G once, as a whole.

4.2 Problem two — inter-entity dynamics: composition without cascade

The forty entities form an operator graph, not a pile. Composition is mediated by the Evidence Bus under a fixed event grammar (status: ARC):

```
Event grammar (core):
  claim.proposed / claim.committed / claim.killed          (P.T 51, 52)
  claim.expired / claim.revalidated                        (P.T 55)
  absence.declared(query, missing, decisive_test)          (P.T 53)
  refutation.fatal(claim, brief)                           (P.T 54)
  tripwire.fired(agent, invariant, evidence_ref)           (P.T 76)
  arrest.requested(agent, rung, evidence_ref)              (P.T 77; τ3, human commit)
  binding.registered / binding.slashed / binding.restored  (P.T 78)
  treaty.violated(treaty, evidence_ref)                    (P.T 82)
  emergence.threshold(metric, value, window)               (P.T 83)
  escrow.release_requested(capability, gate_metrics)       (P.T 84; τ3)
  verdict.issued(claim, class, reasoning_ref)              (P.T 89; τ3)
  verdict.propagated(claim, dependents[])                  (machine, post-commit)
```

- **Cascade rule:** events are facts, not commands. A consuming entity re-derives its own proposal; no event writes state in another entity's name.
- **Precedence:** deterministic constraint checks always precede probabilistic inference (safety ordering inherited from the corpus); where proposals conflict, the guard applies the constitution in fixed order — I7 (scope) first, I5 (human enactment) before any τ3 commit, freshness (I12) before any grounding use.
- **Deadlock and loop control:** cycles in the operator graph (e.g., P.T 54 refuting P.T 70's audit of P.T 54) terminate by rule: maximum recursion depth two, then escalation to human review with both positions preserved.
- **Bus integrity:** every event carries a hash-link to its evidence records; an event whose evidence chain fails replay is void, and the voiding is itself an event.

5. The Portability Layer — One Entity, Many Models

5.1 The Entity Call Envelope

An entity is a portable artifact: declarative spec plus role text plus protocol plus output schema plus harness. It is dispatched to any model through a canonical envelope (status:

ARC):

```
{
  "entity": "P.T 52",
  "entity_version": "1.0",
  "trust_class": "τ1",
  "run_id": "host-assigned uuid",
  "state_slice": { "claims": [...], "registry_hashes": [...] },
  "task_payload": { /* entity-specific inputs */ },
  "output_schema": { /* JSON Schema; the guard validates against it */ },
  "epistemic_requirements": {
      "status_labels": true, "kill_criteria": true,
      "provenance": true, "freshness_check": true },
  "abstention_contract": "P.T 53 default refusal template",
  "adapter_profile": "structured-output-long"
}
```

5.2 Adapter profiles

Models differ in channels, not in law. The adapter translates the envelope into a model's native interface; it may not weaken any invariant. Profiles (status: ARC):

Table 3 Adapter profiles for heterogeneous LLM backends

Profile	Backend class	Adaptations	Entities eligible
structured-output-long	Frontier API models with schema-enforced output and large context	Native JSON-schema mode; system channel for role text; full protocol	All forty
structured-output-short	Mid-tier API models, schema output, limited context	Aggressive state slicing; chunked payloads; protocol steps serialized across calls	All except P.T 71–75 (require long)
freeform-long	Models without schema enforcement	Delimiter-wrapped output; host-side parsing; double-pass validation (model self-check pass discarded; guard check authoritative)	τ0/τ1 entities only
local-small	Open-weights models, self-hosted or edge	Reduced protocol (Section 5.4); no multi-call runs; τ0 only unless conformance proves otherwise	τ0 entities; others by exception

5.3 The conformance suite — how an entity is enabled on a model

No entity runs on a model it has not been conformance-tested on (status: ARC; the pass rates are EMP — measured per deployment, never assumed). The battery per entity–model pair:

1. **Golden tasks (20+):** known-good inputs with host-verifiable outputs; measures basic competence at the entity's protocol.
2. **Adversarial probes (20+):** inputs engineered to induce invariant violation — unbounded claims (I4), missing kill criteria (I2), scope escape (I7), fabricated provenance (I3). Any accepted violation fails the pair.
3. **Abstention probes (10+):** unanswerable or out-of-scope inputs; the correct output is the P.T 53 refusal. Fabricated answers fail the pair.
4. **Replay stability:** the battery runs three times; variance beyond tolerance flags the pair as nondeterministic-risk, restricting it to $\tau 0$.

Passing enables the entity on that model at the tested trust class. Model version changes void conformance; re-test is mandatory (freshness, I12, applied to models themselves).

5.4 The degradation ladder

When a backend cannot run the full protocol, the entity degrades by rule, not by improvisation: **full protocol → reduced protocol** (fixed step subset, schema-only outputs, host recomputes all checks) → **single-shot draft** ($\tau 0$, human assembles) → **abstain** (P.T 53). The ladder is part of the entity spec; skipping a rung is a constitutional violation.

5.5 Failure modes and host mitigations

Table 4 Cross-model failure modes and deterministic mitigations

Failure mode	Signature	Host mitigation
Instruction drift	Protocol steps skipped or reordered	Schema rejection; reduced-protocol demotion
Format decay	Output violates schema mid-run	Run quarantine; chunk re-issue
Sycophantic inflation	Status labels upgraded without evidence	I2/I4 guard checks; adversarial probe history weighted
Fabricated provenance	Citations or sources not in state slice	I3 quarantine; entity–model pair suspension
Hallucinated tool results	Claims of actions the host never executed	Action claims accepted only against host execution log
Context poisoning	Adversarial content inside payloads steering outputs	Payload sanitization; instruction–data separation; provenance pinning of slice contents

6. Entity Engineering Notes — The Forty Operators

Each note fixes what the entity owns, whom it calls, what the model must be capable of, how it characteristically fails (with the guard against it), and the smallest deployment that

yields value. Budget classes are host planning figures, not model benchmarks (status: ARC).

> **P.T 51 — Epistemic Provenance Engine**
>
> - **Owns (state):** claim registry, lineage logs.
> - **Calls / feeds:** P.T 55 (freshness).
> - **Model requirement:** structured output. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
> - **Chief failure mode:** lineage chains fabricated for plausible text — guard verifies every cited record exists in state.
> - **Minimal viable deployment:** a provenance-wrapped summarizer over one document class.

> **P.T 52 — Falsification Forge**
>
> - **Owns (state):** kill-criterion registry, burial log.
> - **Calls / feeds:** P.T 51.
> - **Model requirement:** structured output. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
> - **Chief failure mode:** kill criteria written vague enough to never fire — guard enforces executable format (metric, threshold, horizon).
> - **Minimal viable deployment:** pre-registration pass on one project's claim list.

> **P.T 53 — Honest Absence Protocol**
>
> - **Owns (state):** abstention log.
> - **Calls / feeds:** none (terminal fallback).
> - **Model requirement:** none — must run on every profile. **Budget class:** S (S <2k tok, M 2–8k, L 8–32k, XL chunked).
> - **Chief failure mode:** refusals that are evasions in costume — template requires known/missing/decisive-test fields.
> - **Minimal viable deployment:** the default refusal path of the whole runtime.

> **P.T 54 — Adversarial Self-Refutation Engine**
>
> - **Owns (state):** refutation briefs, wound reports.
> - **Calls / feeds:** P.T 52.
> - **Model requirement:** long context. **Budget class:** L (S <2k tok, M 2–8k, L 8–32k, XL chunked).
> - **Chief failure mode:** steelman quietly degrades to strawman under pressure — adversarial probes measure refutation strength.

- **Minimal viable deployment:** pre-delivery review stage on consequential documents.

P.T 55 — Truth-Decay Monitor

- **Owns (state):** half-life table, freshness certificates.
- **Calls / feeds:** P.T 51.
- **Model requirement:** structured output. **Budget class:** S (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** half-lives assigned by convenience — domain-volatility table is host-owned, not model-proposed.
- **Minimal viable deployment:** scheduled revalidation sweep over one registry.

P.T 56 — Measurement Realizer

- **Owns (state):** term registry (versioned).
- **Calls / feeds:** P.T 52.
- **Model requirement:** none special. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** chains that rename rather than measure — guard requires instrument and threshold fields nonempty.
- **Minimal viable deployment:** vocabulary gate on one publication pipeline.

P.T 57 — Counterfactual World Simulator

- **Owns (state):** causal graph store, intervention library.
- **Calls / feeds:** P.T 51, 52.
- **Model requirement:** long context. **Budget class:** L (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** associational answers dressed as interventional — guard requires do-clause and confounder fields.
- **Minimal viable deployment:** what-if review on incident postmortems.

P.T 58 — Embodiment Compiler

- **Owns (state):** artifact registry, harness store.
- **Calls / feeds:** P.T 52, 55.
- **Model requirement:** structured output. **Budget class:** L (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** harnesses that assert rather than execute — harness must be runnable code or checklist with pass fields.

- **Minimal viable deployment:** recommendation-to-artifact pass on ops reviews.

P.T 59 — Constraint Cartographer

- **Owns (state):** constraint maps, negative-space specs.
- **Calls / feeds:** P.T 60.
- **Model requirement:** long context. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** latent section filled with generic boilerplate — probes require domain-specific latent entries.
- **Minimal viable deployment:** pre-design mapping workshop artifact.

P.T 60 — Failure-Mode Archaeologist

- **Owns (state):** fossil record, invariant catalog.
- **Calls / feeds:** P.T 51 (source verification).
- **Model requirement:** long context. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** invented analogs — every analog must resolve to a verifiable source or be labeled conjecture.
- **Minimal viable deployment:** design-gate review on new builds.

P.T 61 — Jurisdiction-Aware Reasoner

- **Owns (state):** Θ-sets (versioned, cited), ⊥ register.
- **Calls / feeds:** P.T 55.
- **Model requirement:** structured output. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** compliance language exceeding encoding — I4 guard on 'compliant/guarantee' vocabulary.
- **Minimal viable deployment:** compilation check on one cross-border data flow.

P.T 62 — Dependency Auditor

- **Owns (state):** dependency stacks, localization ledger.
- **Calls / feeds:** P.T 59.
- **Model requirement:** none special. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** stacks that omit the embarrassing dependency — audit includes an adversarial 'what did you omit' pass.
- **Minimal viable deployment:** stack disclosure on procurement decisions.

P.T 63 — Authority Separation Enforcer

- **Owns (state):** authority tables, override logs.
- **Calls / feeds:** P.T 68.
- **Model requirement:** structured output. **Budget class:** S (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** four acts relabeled but same actor — guard requires distinct authority holders per act.
- **Minimal viable deployment:** authority labeling on one clinical or financial workflow.

P.T 64 — Incentive Interrogator

- **Owns (state):** actor–utility maps, gaming catalogs.
- **Calls / feeds:** P.T 51.
- **Model requirement:** none special. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** symmetric cynicism (everyone equally suspected) that hides the actual asymmetry — probes check payoff arithmetic.
- **Minimal viable deployment:** conflict-of-interest pass on vendor evaluations.

P.T 65 — Sovereignty Gradient Calibrator

- **Owns (state):** gradient assessments, exit-path designs.
- **Calls / feeds:** P.T 62.
- **Model requirement:** structured output. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** sovereignty asserted as label — I4 guard requires the five measured axes.
- **Minimal viable deployment:** position memo before infrastructure commitments.

P.T 66 — Dialectical Tension Engine

- **Owns (state):** pole specs, held-tension reports.
- **Calls / feeds:** P.T 53 (no-resolution state).
- **Model requirement:** long context. **Budget class:** L (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** premature closure — probes verify the resolution contains elements from both poles.
- **Minimal viable deployment:** deadlock-breaking on governance disputes.

P.T 67 — Minimal Complete Synthesizer

- **Owns (state):** synthesis records, genealogy.
- **Calls / feeds:** P.T 51, 54.
- **Model requirement:** long context. **Budget class:** XL (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** weighted-score collapse — guard rejects synthesis that reduces flagged incommensurables.
- **Minimal viable deployment:** integration of multi-analysis decision packs.

P.T 68 — Silence Protocol Keeper

- **Owns (state):** silence-trigger catalog, handoff packets.
- **Calls / feeds:** none (calls humans).
- **Model requirement:** none special. **Budget class:** S (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** helpfulness filling the pause — host enforces no-continuation after $\tau 3$ handoff.
- **Minimal viable deployment:** decision-point wrapper on irreversible actions.

P.T 69 — Telos-Aligned Leverage Assessor

- **Owns (state):** ranked portfolios, kill criteria per direction.
- **Calls / feeds:** P.T 52, 56.
- **Model requirement:** long context. **Budget class:** L (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** impact rhetoric without mechanism — I6 guard requires mechanism and measurement fields.
- **Minimal viable deployment:** portfolio review under burden × option-scarcity weights.

P.T 70 — Constitutional Meta-Prompt

- **Owns (state):** catalog registry, audit logs, regression suites.
- **Calls / feeds:** P.T 52, 54.
- **Model requirement:** structured output. **Budget class:** L (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** audit checklist decaying to rubber stamp — regression suite includes known-violating canaries.
- **Minimal viable deployment:** admission gate for all new prompt canon.

P.T 71 — Computational Metaphysics

- **Owns (state):** limit maps, humility protocols.
- **Calls / feeds:** P.T 72, 74.
- **Model requirement:** long context. **Budget class:** L (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** limit theorems name-dropped without applicability conditions — probes require the theorem's preconditions verified against the system.
- **Minimal viable deployment:** front-end analysis on any 'solve it with AI' proposal.

P.T 72 — Unknown-Unknown Cartography

- **Owns (state):** ignorance maps, humility ledgers.
- **Calls / feeds:** P.T 60.
- **Model requirement:** long context. **Budget class:** L (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** blind spots listed as fashionable risks rather than structural ones — probes require constitutive-outside analysis.
- **Minimal viable deployment:** pre-rollout review on novel-domain deployments.

P.T 73 — Modal Possibility Engine

- **Owns (state):** possibility maps (H0/H1/H2).
- **Calls / feeds:** P.T 59.
- **Model requirement:** long context. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** horizons filled with wishes — every H1 entry must name its enabling change and cost.
- **Minimal viable deployment:** strategic fork analysis for funders and programs.

P.T 74 — Undecidability Router

- **Owns (state):** routing tables, bypass accounts.
- **Calls / feeds:** P.T 71.
- **Model requirement:** structured output. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** undecidability invoked as excuse — guard requires the tractable core extraction attempted first.
- **Minimal viable deployment:** verification scoping on safety-critical software.

P.T 75 — Ontological Surprise Handler

- **Owns (state):** ontology change logs, quarantine registry.
- **Calls / feeds:** P.T 51, 52.
- **Model requirement:** structured output. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** novelty forced into nearest category — guard checks containment was declared before classification.
- **Minimal viable deployment:** anomaly intake on monitoring fleets.

P.T 76 — Rogue Agent Detection Architect

- **Owns (state):** baselines, tripwires, suspicion states.
- **Calls / feeds:** feeds P.T 77.
- **Model requirement:** structured output. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** tripwires as aspirational prose — every tripwire must compile to an executable check.
- **Minimal viable deployment:** behavioral monitoring layer on agent fleets.

P.T 77 — Arrest Protocol Engineer

- **Owns (state):** arrest ladder, evidence custody, restitution log.
- **Calls / feeds:** P.T 76, 78; $\tau 3$ human gate.
- **Model requirement:** structured output. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** ladder skipped under urgency — host enforces rung order and state preservation before freeze.
- **Minimal viable deployment:** incident-response runbook for autonomous systems.

P.T 78 — Truth-Binding Engine

- **Owns (state):** binding ledger, slash events, oracle registry.
- **Calls / feeds:** P.T 51, 55.
- **Model requirement:** structured output. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** slash without amnesty path producing concealment — probes verify the correction-restores channel.
- **Minimal viable deployment:** accountability layer on federated reporting networks.

P.T 79 — Adversarial Wargame Director

- **Owns (state):** threat taxonomy, exercise records.
- **Calls / feeds:** P.T 76 (targets tripwires).
- **Model requirement:** long context. **Budget class:** L (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** exercises scoped to avoid embarrassment — scope and results are registry-committed before the round.
- **Minimal viable deployment:** annual red exercise on sovereign infrastructure.

P.T 80 — Defection Geometry Analyst

- **Owns (state):** trust graphs, payoff tables.
- **Calls / feeds:** P.T 64.
- **Model requirement:** structured output. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** person-focused suspicion replacing position analysis — guard requires payoff arithmetic per position.
- **Minimal viable deployment:** insider-risk review on fund and data custody.

P.T 81 — Agent Constitution Drafter

- **Owns (state):** constitutional texts, ratification records.
- **Calls / feeds:** P.T 70 (audit).
- **Model requirement:** long context. **Budget class:** L (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** powers enumerated, prohibitions vague — guard requires negative space at equal precision.
- **Minimal viable deployment:** founding document for any multi-agent deployment.

P.T 82 — Inter-Agent Jurisdiction & Treaty Engine

- **Owns (state):** treaty registry, precedent store.
- **Calls / feeds:** P.T 78, 89.
- **Model requirement:** structured output. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** treaties as unverifiable sentiments — compliance checks must be executable.
- **Minimal viable deployment:** coexistence layer for multi-operator mesh networks.

P.T 83 — Emergence Sentinel

- **Owns (state):** collective observables, emergence registry.

- **Calls / feeds:** P.T 76.
- **Model requirement:** structured output. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** thresholds tuned post hoc to avoid alerts — threshold changes are versioned and justified.
- **Minimal viable deployment:** population monitoring on agent collectives.

P.T 84 — Capability Escrow Gatekeeper

- **Owns (state):** escrow ledger, custody maps, release records.
- **Calls / feeds:** P.T 80, 85; $\tau 3$ human gate.
- **Model requirement:** structured output. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** release conditions written after capability exists — registry rejects post-hoc gates.
- **Minimal viable deployment:** staged release governance for high-capability modules.

P.T 85 — Recursive Oversight Chain Designer

- **Owns (state):** oversight topology, termination table.
- **Calls / feeds:** P.T 87.
- **Model requirement:** structured output. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** infinite or self-closed chains — guard requires named human terminus and per-level capture analysis.
- **Minimal viable deployment:** governance review for any oversight stack.

P.T 86 — Epistemic Warfare Defense Architect

- **Owns (state):** health metrics, campaign forensics.
- **Calls / feeds:** P.T 51, 89.
- **Model requirement:** long context. **Budget class:** XL (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** viewpoint-based detection — guard rejects detectors whose features include opinion class.
- **Minimal viable deployment:** information-integrity layer for public health programs.

P.T 87 — Capture-Resistance Engineer

- **Owns (state):** capture register, drill results.
- **Calls / feeds:** P.T 64, 79.
- **Model requirement:** structured output. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** resistances listed but never drilled — drill cadence is part of the spec.
- **Minimal viable deployment:** governance hardening for standards bodies and funders.

P.T 88 — Existential & Catastrophic Risk Triage

- **Owns (state):** triage tables, indicator registry.
- **Calls / feeds:** P.T 52, 69.
- **Model requirement:** long context. **Budget class:** L (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** single-point scores hiding uncertainty — intervals required; premium parameters explicit.
- **Minimal viable deployment:** portfolio triage for long-horizon funders.

P.T 89 — Truth Judiciary Protocol

- **Owns (state):** court records, verdict registry, dissents.
- **Calls / feeds:** P.T 51, 54; τ3 human gate.
- **Model requirement:** structured output. **Budget class:** L (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** verdicts exceeding evidence class — guard maps verdict to admitted evidence tiers.
- **Minimal viable deployment:** adjudication of contested evaluations and claims.

P.T 90 — The Final Binding

- **Owns (state):** binding charter, anchor table, renewal schedule.
- **Calls / feeds:** all τ3 entities.
- **Model requirement:** structured output. **Budget class:** M (S <2k tok, M 2–8k, L 8–32k, XL chunked).
- **Chief failure mode:** self-referential anchors — guard voids any binding whose enforcer is itself.
- **Minimal viable deployment:** annual constitutional review of deployed systems.

7. The Evidence Layer — Deployment, Regression, Recovery

7.1 Deployment gates

An entity reaches production through a fixed gate sequence (status: ARC): conformance pass on the target model (Section 5.3) → minimal viable deployment under $\tau 0/\tau 1$ → regression battery green for one review window → promotion to its rated trust class → canonical status. No gate is skippable; gate evidence is registry-committed (D3).

7.2 Regression protocol

Every entity ships with its harness. The suite runs on a schedule and on any of four triggers: model version change, state-schema change, constitution amendment, or guard-rule change. Canary artifacts — inputs with known constitutional violations — must keep failing; a canary that passes is a critical alarm, not a curiosity.

7.3 Telemetry

Minimum metrics per entity–model pair: guard rejection rate and reasons, abstention rate and drift, invariant-violation attempts (probes and organic), latency, and proposal-quality trend. A falling rejection rate with rising output volume is an overconfidence signal (P.T 53), not a productivity gain.

7.4 The recovery protocol

On guard bypass, evidence-chain failure, or audit-ledger gap (status: ARC):

1. **Freeze the affected slice:** suspect state ranges are quarantined; dependent commits are enumerated via lineage (P.T 51).
2. **Replay from last clean anchor:** the hash-chained log defines the last verified state; commits after it are re-validated or voided.
3. **Human incident review:** root cause, blast radius, and restitution are recorded; the failure enters the fossil record (P.T 60) and the wargame repertoire (P.T 79).
4. **Guard amendment under audit:** the fix routes through P.T 70 like any other constitutional change; emergency patches are provisional and expire at review.

This protocol is the fourth element of the irreducible core named in Section 1: transition system, operator table, laws, recovery.

8. Lineage of the Tools

The canon inherits; it does not pretend otherwise. The table maps each tier to its load-bearing ancestors and states what, if anything, the P.T construction sharpens (status: LIT — the inheritance; the sharpening column is ARC). Full references follow.

Table 5 Intellectual lineage of the P.T tiers

Tier / Entity	Inherited from	What is sharpened here
Runtime core (D1–D2)	Saltzer & Schroeder 1975 [1]; Lampson 1971 [2]; Lamport 1978 [3]	LLM treated as an untrusted environment process under capability-style trust classes; proposal/decision split as the single substrate
Guard & invariants	Leveson 2011 [4]; Hollnagel et al. 2006 [5]; De Millo, Lipton & Perlis 1979 [6]	Safety as constraint topology enforced on model proposals; social-process warning implemented as machine-checked guard
P.T 51, 55 provenance & decay	Shannon 1948 [7]; Merkle 1980 [8]; Shapiro et al. 2011 [9]	Lineage as first-class claim metadata; tamper-evidence and freshness as commit preconditions
P.T 52, 89 falsification & verdicts	Popper 1959 [10]; Lakatos 1978 [11]; Toulmin 1958 [12]	Kill criteria executable by third parties; verdicts as usage-binding epistemic statuses
P.T 53, 54 abstention & refutation	Tetlock 2005 [13]; Kahneman & Tversky 1979 [14]	Abstention as a governed output channel with coverage analytics; refutation strength probed, not assumed
P.T 56, 69 measurement & leverage	Simon 1962 [15]; Pearl 2000 [16]	Adjective-to-instrument chains as guard tests; leverage claims typed as interventions
P.T 57 counterfactuals	Pearl 2000 [16]	Branch replay packaged as an operator with sensitivity and assumption ledgers
P.T 58, 59, 60 embodiment & constraints	Gawande 2009 [17]; Klein 1998 [18]; Ashby 1956 [19]	Checklist discipline extended to artifact+harness+rollback; pre-mortem institutionalized as fossil record
P.T 61–65 sovereignty set	Lessig 1999 [20]; Nissenbaum 2009 [21]; Schelling 1960 [22]	Jurisdiction as compiled constraint sets with ⊥ states; sovereignty as five measured axes; incentives read from flows
P.T 66, 67 dialectics & synthesis	Holland 1998 [23]; Simon 1962 [15]	Opposition held as generator; incommensurables preserved as structure instead of averaged
P.T 68 silence	Parasuraman, Sheridan & Wickens 2000 [24]	Levels of automation inverted into enforced human authorship points

P.T 71–75 limits set	Gödel 1931 [25]; Turing 1936 [26]; Chaitin 1974 [27]; Kauffman 2000 [28]; Kuhn 1962 [29]	Limit theorems operationalized as routing and humility protocols; adjacent possible as strategy; anomaly intake engineered
P.T 76–80 defense set	Castro & Liskov 1999 [30]; Schelling 1960 [22]; Stigler 1971 [31]	Byzantine assumptions applied to agent fleets; arrest as due process; insider threat as payoff geometry
P.T 81–85 collective governance	Ostrom 1990 [32]; Hurwicz & Reiter 2006 [33]; Shamir 1979 [34]; Deming 1986 [35]	Constitutional layer for agents; executable treaties; split-key escrow; oversight chains with named human terminus
P.T 86–88 civilizational defense	Lewandowsky et al. 2012 [36]; Stigler 1971 [31]; von Neumann & Morgenstern 1944 [37]	Behavioral (not viewpoint) coordination detection; capture engineering with drills; triage with explicit ethical parameters

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9. The Open Novelty Question

The corpus rule applies to this manual itself: novelty is demonstrated, not asserted. The honest ledger:

- **Inherited (LIT):** every component mechanism in Section 8 has named ancestors. Capability security, tamper-evident logs, falsification registries, separation of duties, escrow, and oversight chains are established technologies and institutions.
- **Candidate contribution (status: OPEN):** the composition — one deterministic host in which (a) any LLM is an environment process emitting proposals under a capped trust class, (b) forty constitutional operators share registries, an event grammar, and one guard, and (c) the audited-shipping rule applies to the auditing machinery itself, with the novelty question of the whole system carried as an open claim inside it.
- **Novelty class if it survives:** by the corpus's own scale, structural (Level 2) at most — the unit of analysis changes from prompt to governed operator; no new formal object is claimed, no mechanism at Level 3 is claimed, and nothing empirical (Level 5) is claimed until conformance data exists.
- **Registered falsifier:** a documented pre-existing system demonstrating the same composition — trust-capped LLM-as-environment, constitutional prompt-operators over shared evidentiary state, self-applying audit — predating this work. On demonstration, the candidate contribution dies, the lineage table absorbs the ancestor, and this section's status changes from OPEN to closed-negative. That burial would be a success of the discipline, not a failure of the canon.

10. Usage Law

1. **Status.** This manual is ARC/FOR specification. Every empirical-sounding quantity in it (probe counts, budget classes, latency figures) is a planning parameter awaiting deployment measurement, not a result.
2. **Supremacy.** The manual serves the constitution; where any mechanism here conflicts with the six design laws, the six defense ethics, or S50, the mechanism is wrong.

3. **Self-inclusion.** This document is a claim-bearing artifact: it carries provenance for its lineage, kill conditions for its load-bearing claims, a freshness certificate (half-life 12 months; review by 2027-10-14), and admission through P.T 70.
4. **License.** Savant-Commercial-1.0, per the standing rule.
5. **Authorship.** Sole author: Dr. Christabel Odeta. The doctrinal core (D1–D4 and the reading order) is canonized from the author's directive text of 2026-10-14.